Hyperspectral Image Classification Based on Subgraph-Dependent Neural Network

Yun Ding, *Member, IEEE*, Chao Li, Fulin Luo, *Senior Member, IEEE*, and Chunhou Zheng, *Member, IEEE*

***Abstract*— Hyperspectral image classification methods based on subgraph neural networks (SGNNs) are rarely explored, and its advantage is that it can alleviate the neighbor explosion problem. After applying SGNNs to hyperspectral image (HSI) classification, the imbalanced topology structure in the internal subgraph leads to poor classification performance due to the intraclass and interclass spectral feature variation. Based on this, we proposed a novel subgraph-dependent neural network (SGDNet) for HSI classification. Specifically, we first segmented the large graph into a series of subgraphs and proposed a subgraph-dependent convolution method for imbalanced subgraph structures to achieve effective feature smoothing within each subgraph. It mainly utilized the degree feature embedding with the residual to improve the feature diversity, as well as the intraclass distance of clustering measurement to determine the optimal subgraph convolution layers in a feedback way. Second, we developed the strategy of the long-range dependency to address the inevitable local structure dependence among nodes within subgraphs. It utilized an anchor point-based position coding method to capture the relative positions of unlabeled nodes in relation to all labeled nodes within the graph and further constructed the structural loss function for effective training, achieving subgraph optimization. Comprehensive experiments demonstrated the superior performance of the SGDNet model on three publicly available HSI datasets, compared to other popular methods. The source code will be available at https://github.com/lichao226211/SGDNet.**

***Index Terms*— Adaptive polymerization, graph convolutional networks (GCNs), hyperspectral image (HSI), subgraph optimization.**

I. Introduction

HYPERSPECTRAL imaging is a significant technique in remote sensing that utilizes hyperspectral imagers to capture data across numerous spectral bands [1], [2]. These imagers cover a range from visible to near-infrared wavelengths (approximately 400–2500 nm), providing more detailed spectral information compared to conventional RGB [3] and multispectral images [4]. Hyperspectral image (HSI) classification is to assign labels to individual pixels based on their distinct spectral characteristics. However, two key challenges affect the classification accuracy of individual pixels in HSIs. The first challenge stems from the spectral variability present within ground objects [5], which can be influenced by factors like lighting and atmospheric conditions. This variability exacerbates the differences within the same class, making accurate classification be a complex task. The second challenge arises from the high expenses associated with acquiring ground-truth labels [6], [7], leading to a scarcity of training samples. Nonetheless, recent advancements in deep learning present promising solutions to these challenges by integrating limited training samples with deep learning techniques. Currently, three predominant approaches are prevalent in HSI classification: convolutional neural networks (CNNs), Transformers, and graph convolutional networks (GCNs).

CNNs, as a representative of deep learning [8], [9], [10], [11], leverage convolution and pooling to exhibit superior performance in tasks such as HSI classification [12], [13], [14] and object detection [15]. In order to improve the performance of the HSI classification, the researchers employed 3-D CNNs [16], [17], [18], [19], [20] to combine local spatial and spectral characteristics and implemented regularization techniques to mitigate overfitting issues. Despite the effectiveness of existing CNN-based methods in capturing spectral and spatial information of HSI, their classification performance is somewhat limited when dealing with long-range information extraction. To address this issue, the Transformer models utilize the self-attention mechanism to capture long-range dependencies between HSI pixels. For instance, researchers [21], [22] improved the classification performance by combining spectral–spatial morphological convolutional operations or absolute position encoding with the self-attention mechanism. However, the dense connection relationships of self-attention mechanisms are easy to be influenced by large spectral variability, leading to confusion in semantic information. GCNs are able to extract irregular region information by using a flexible sparse graph structure and a message-passing mechanism, and achieve information aggregation from structural neighbors. Recently, some researchers have proposed integrating intrinsic spectral details with spatial information for effective HSI classification [23], [24], [25], [26].

In fact, GCN-based algorithms are widely used in HSIs for several reasons. First, spectral variability is ubiquitous in HSIs. Employing a flexible graph structure in graph convolution is able to aggregate neighboring spectral feature information, achieving effective feature smoothing and alleviating

Received 11 February 2025; revised 15 April 2025; accepted 17 May 2025. Date of publication 23 May 2025; date of current version 3 June 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62302002 and Grant 62371076, in part by the Key Program of Natural Science Project of Educational Commission of Anhui Province under Grant 2023AH050080, and in part by the New Chongqing Youth Innovative Talents Project under Grant CSTB2024NSCQQCXMX0071. *(Corresponding authors: Fulin Luo; Chunhou Zheng.)*

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Digital Object Identifier 10.1109/TGRS.2025.3573046

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the spectral variability phenomenon [27]. Second, the spatial resolution of HSIs is generally low, and their spatial structure is complex. Using an irregular graph structure can better characterize the complex spatial distribution of neighborhoods and structural adjacency relationships. Third, the acquisition of training samples in HSIs is extremely costly, leading to a general scarcity of training samples. GCNs, as an effective semi-supervised learning paradigm, are able to establish the connectivity relationships between a small number of training samples and a large number of testing samples. It could fully pass prior knowledge to the testing samples via a message-passing mechanism, achieving favorable classification results even with a limited number of training samples. For example, the spatial–spectral adaptive probabilistic GCN (SSAPGCN) [28] adaptively learns local graph structures and output features and provides feedback to graph learning. Generally, the output features of these GCN-based methods come from its neighbors in the previous layer. Clearly, the deeper we backtrack, the more multihop neighbors to support the computation of the root nodes [2]. The number of support nodes potentially grows exponentially with the GCN depth. In other words, they still face a formidable challenge that is the neighbor explosion due to their message-passing mechanism.

To tackle this issue of neighbor explosion, researchers proposed the mini-batch sampling [29], [30] and subgraph-based sampling methods to address the neighbor explosion issue. For example, Hong et al. [31] used a mini-batch GCN (miniGCN) training approach to aggregate a fixed number of neighbors. They then integrated CNN and GCN modules to form the FuNet-C method. Additionally, Chiang et al. [32] introduced an effective deep and large-scale GCN training algorithm called Cluster GCN, which constructs dense subgraphs through interconnected node sampling based on graph clustering structures. Compared to mini-batch node sampling methods, subgraph-based methods [subgraph neural networks (SGNNs)] have the following advantages. First, for large-scale HSI, miniGCN-based methods sample a fixed size of neighborhood during back-propagation, but its overhead is still large when the message-passing paradigm goes deep. However, the subgraph-based methods sample a block of nodes that are associated with a subgraph and restrict the neighborhood search within this subgraph, reducing its computational overhead. Second, classification methods based on minibatch sampling often encounter the challenge of being influenced by the similarity of spectral feature variation between classes during the node sampling process. However, the subgraph-based methods, due to their ability to ensure homogeneous structural information among nodes, can achieve effective intraclass information aggregation, alleviating the influence of spectral variability to a certain extent. Third, the subgraphs enable direct focus on regions of interest in HSI without being restricted by pre-defined sampled neighbors, while still retaining the capacity to capture certain neighboring structure information, preventing structural information loss.

Despite the fact that SGNN-based methods are able to preserve structural information between nodes and reduce information loss, the above-mentioned SGNN-based methods are still facing a serious limitation that they aggregate the fixed K-order neighbors' information for each subgraph structure in the message-passing paradigm. In fact, they ignore the imbalanced topological structures of different subgraphs due to the influence of spectral variability, which leads to the varying degrees of connectivity. Using the aggregation strategy of fixed K-order neighbors' information with an imbalanced subgraph structure in the message-passing mechanism makes it difficult to achieve effective smoothing of feature information within subgraphs. In order to address this issue, we proposed a novel subgraph-dependent neural network (SGDNet) method that can adaptively identify and capture the neighbor orders of each subgraph, achieving flexible feature information aggregation and good classification results. We designed a subgraph-dependent convolution method for imbalanced subgraph structure developed a strategy of structural long-range dependency to capture the relative structural position information within the graph. Next, we further constructed the structural loss function for effective training. Comprehensive experiments on three popular HSI datasets, including Indian Pines, Houston, and PaviaU, demonstrated the superior performance of the SGDNet model. The main contributions of this article can be summarized as follows.

1) This article proposed a subgraph-dependent convolution method for imbalanced subgraph structure, which mainly utilized the degree feature embedding with the residual to improve the feature diversity and the intraclass distance of clustering to guarantee the optimal subgraph convolution layers in a feedback way, achieving effective feature smoothing within subgraphs.
2) Considering that the subgraph inevitably disrupting global dependencies among nodes, we developed the strategy of structural long-range dependency to address the above issues, which utilized an anchor point-based position coding method to capture the relative positions of unlabeled nodes in relation to all labeled nodes within the graph and constructed an structural loss function, mitigating the influence of imbalances in subgraph structures.
3) We proposed an efficient SGDNet method for effective HSI classification by addressing the imbalance of subgraph structure and neighbor explosion of GCNs, which mainly consisted of the fusion of subgraph-dependent convolution and subgraph optimization. Comprehensive experiments demonstrate that the SGDNet achieved better classification results compared with the other popular methods.

The remaining sections of this article are presented as follows. Section II provides an introduction to GCN. In Section III, the SGDNet method is described in detail. Section IV presents the experimental setup and methodology employed in this article. Finally, Section V concludes the article by summarizing the findings and contributions.

II. Related Work

In this section, we systematically analyze the basic definition of GCN and introduce the principle of Cluster-GCN because they are close to the proposed SGDNet.

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A. Cluster-GCN

Cluster-GCN is a technique that enables deep training and handling of large-scale graphs, offering improvements over GCN. In Cluster-GCN, given a graph $\boldsymbol{G}$, it is initially partitioned into graph segments. These segments can be represented as n graphs, denoted as $\boldsymbol{V} = [\boldsymbol{V}_1, \ldots \boldsymbol{V}_n]$, where $\boldsymbol{V}_i$ represents the nodes included in the ith subgraph, and $\boldsymbol{E} = [\boldsymbol{E}_1, \ldots \boldsymbol{E}_n]$, where $\boldsymbol{E}_i$ represents the edge set included in the ith subgraph. Thus, the original graph $\boldsymbol{G}$ can be expressed as follows:

$$\overline{\boldsymbol{G}} = [\boldsymbol{G}_1, \ldots, \boldsymbol{G}_n] = [\{\boldsymbol{V}_1, \boldsymbol{E}_1\}, \ldots, \{\boldsymbol{V}_n, \boldsymbol{E}_n\}].$$

After reorganizing the nodes, in order to effectively handle the graph structure, the adjacency matrix is partitioned into n^2 submatrices. This division allows for specific operations and computations to be performed on different submatrices. The adjacency matrix can be represented as follows:

$$\boldsymbol{A} = \overline{\boldsymbol{A}} + \triangle = \begin{bmatrix} \boldsymbol{A}_{11} & \ldots & \boldsymbol{A}_{1n} \\ \ldots & \ddots & \ldots \\ \boldsymbol{A}_{n1} & \ldots & \boldsymbol{A}_{nn} \end{bmatrix} \tag{1}$$

and

$$\overline{\boldsymbol{A}} = \begin{bmatrix} \boldsymbol{A}_{11} & \ldots & 0 \\ \ldots & \ddots & \ldots \\ 0 & \ldots & A_{nn} \end{bmatrix}, \triangle = \begin{bmatrix} 0 & \ldots & \boldsymbol{A}_{1n} \\ \ldots & \ddots & \ldots \\ \boldsymbol{A}_{n1} & \ldots & 0 \end{bmatrix} \tag{2}$$

where $\overline{\boldsymbol{A}}$ represents the complete adjacency matrix of the entire large graph after partitioning. The main diagonal element $\boldsymbol{A}_{ii}$ represents the adjacency matrix of each subgraph, with a size of $|\boldsymbol{V}_i| \times |\boldsymbol{V}_i|$, where $|\boldsymbol{V}_i|$ denotes the number of nodes in the ith subgraph. The sub-diagonal element $\boldsymbol{A}_{ij}$ represents the adjacency information between subgraph i and subgraph j. Specifically, $\boldsymbol{A}_{ij}$ captures the adjacency information between the nodes of subgraph i and subgraph j in the original graph.

Finally, the GCN for the entire large graph can be decomposed into a set of GCNs for each subgraph. This means that the GCN model is applied individually to each subgraph, allowing for localized computations and operations within each subgraph. This decomposition enables more efficient processing and handling of the large graph structure, $\overline{\boldsymbol{A}}'$ denotes the normalized version of $\overline{\boldsymbol{A}}$. The final embedding matrix can be obtained as follows:

$$\begin{aligned} \boldsymbol{Z}^{(L)} &= \overline{\boldsymbol{A}}' \sigma\left(\overline{\boldsymbol{A}}' \sigma\left(\cdots \sigma\left(\overline{\boldsymbol{A}}' \boldsymbol{X} \boldsymbol{W}^{(0)}\right) \boldsymbol{W}^{((1)}\right) \cdots\right) \boldsymbol{W}^{((L-1)} \\ &= \begin{bmatrix} \overline{\boldsymbol{A}}'_{11} \sigma\left(\overline{\boldsymbol{A}}'_{11} \sigma\left(\cdots \sigma\left(\overline{\boldsymbol{A}}'_{11} \boldsymbol{X} \boldsymbol{W}^{((0)}\right) \boldsymbol{W}^{(1)}\right) \cdots\right) \boldsymbol{W}^{(L-1)} \\ \vdots \\ \overline{\boldsymbol{A}}'_{nn} \sigma\left(\overline{\boldsymbol{A}}'_{\mathrm{nn}} \sigma\left(\cdots \sigma\left(\overline{\boldsymbol{A}}'_{\mathrm{nn}} \boldsymbol{X} \boldsymbol{W}^{(0)}\right) \boldsymbol{W}^{(1)}\right) \cdots\right) \boldsymbol{W}^{(L-1))} \end{bmatrix}. \end{aligned} \tag{3}$$

The loss function of the corresponding subgraph can be expressed as follows:

$$\mathcal{L}_{\overline{\boldsymbol{A}}'_{ii} = \frac{1}{|\boldsymbol{V}_i|}} \sum_{t \in \boldsymbol{V}_i} \mathrm{loss}\left(\boldsymbol{y}_i, \boldsymbol{z}_i^{(L)}\right). \tag{4}$$

B. GCNs for HSI Classification

In the domain of HSI classification, the rapid development of deep learning technologies has given rise to numerous innovative methods that enhance the accuracy and efficiency of classification tasks. CNNs have become a central tool for processing HSIs because CNNs can automatically learn to identify local features. For instance, the multichannel hybrid 2D-3D-CNN model [33] integrated spectral with spatial information, employing a 3-D convolutional network to process HSI data, thereby significantly improving the precision of HSI classification. However, traditional CNN models exhibit certain limitations in capturing global features. Consequently, Transformers, based on self-attention mechanisms, have a distinct advantage in capturing long-range dependencies. For example, the cross-attention fusion network (CAF-Former) [34] integrated the local feature information from CNNs and the global feature information from Transformers using a cross-attention mechanism for HSI classification, resulting in a notable improvement in classification performance. In addition to the integration of CNNs and Transformers, GCNs have also demonstrated unique advantages in HSI classification due to aggregating information through the message-passing mechanism between neighboring nodes. For example, the multiscale random-shape convolution and adaptive GCN fusion network (MRCAGCFN) [35] adaptively fused multiscale information, enhancing the adaptability to data of different morphologies and improving the robustness of classification. Furthermore, the GCN based on multiscale feature search (MFSGCN) [36] further strengthened the discriminative power of features at different scales. Its automatic weight allocation mechanism allows the model to more flexibly adjust the influence of features according to specific task requirements. Moreover, to overcome the potential insufficiencies of single models, researchers have attempted to integrate various technologies to further improve the accuracy and efficiency of HSI classification. The framework based on multichannel GCNs and class-guided attention mechanism (MG2A) [37] was a typical fusion approach, which adaptively fused the features of multiple channels through a class-guided attention mechanism, enhancing the model's accuracy during the classification process. Liu et al. [38] designed the heterogeneous dual enhanced network (HDECGCN), which combined CNNs and multiscale dynamic GCNs. Through multiscale dynamic tuning, it overcame the accuracy degradation issues that deep convolution layers may cause when dealing with complex scenes. This method significantly improves classification accuracy by enhancing the complementarity between superpixels and pixels and successfully enhances the model's adaptability at different scales.

Although the GCN-based methods have made significant progress in complexly HSIs, they still face a formidable challenge that is neighbor explosion due to their message-passing mechanism. Because GNN-based methods can retain the structural information to a large extent, they are considered as one of the effective methods to solve the neighbor explosion problem. In fact, SGNN-based methods have been widely studied in the field of computer vision and have proven their effectiveness, but the application of SGNN-based methods in HSI classification is rarely explored.

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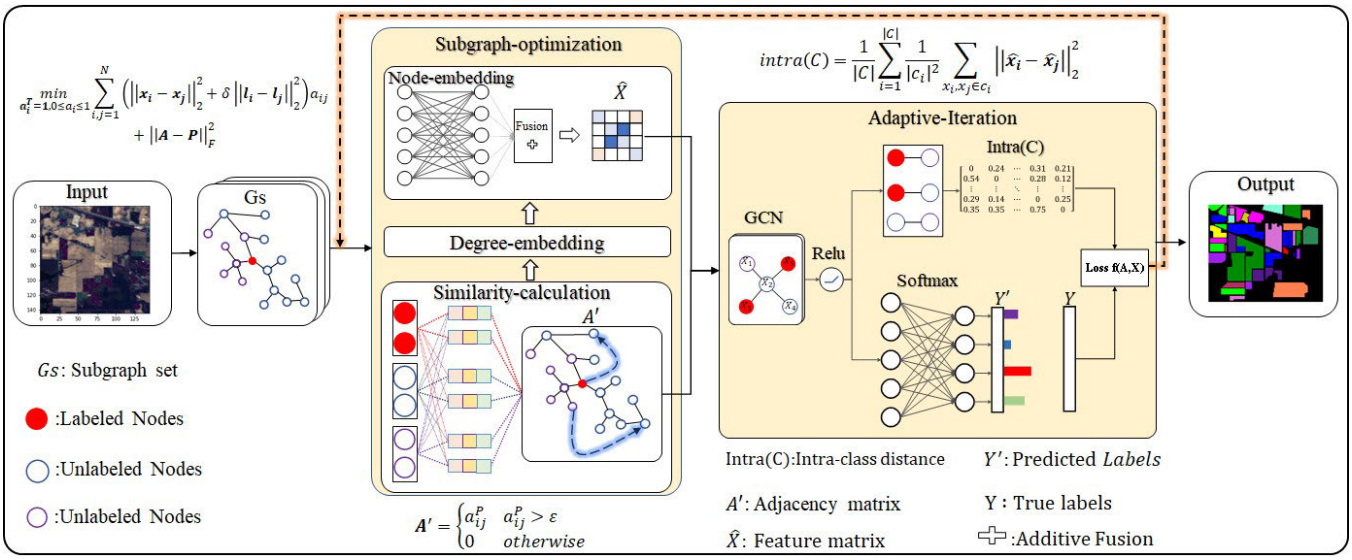


Fig. 1. Algorithmic framework of SGDNet, divided into distinct phases: input, subgraph optimization, adaptive iteration, and output. Within this framework, a dynamic cyclic process occurs that transitions from graph optimization to information aggregation once the criteria for intraclass distance judgment are satisfied.

III. Subgraph-Dependent Neural Network

In this section, we delve into the proposed SGDNet model in detail. The overall workflow of the proposed SGDNet can be vividly observed in Fig. 1, and the SGDNet can be divided into the following four key steps: input, subgraph construction, subgraph-dependent convolution, and subgraph optimization. In the input stage, pixel feature data is processed along with structural information, forming a large graph. An efficient Metis [39] algorithm is employed to effectively partition this large graph into multiple subgraphs, taking into account the topological diversity among them. Next, an adaptive iterative method is proposed to perform message aggregation with an imbalanced topology subgraph structure, which consists of two key components: 1) subgraph-dependent convolution and 2) subgraph optimization. These components are intricately intertwined, forming a tightly knit bidirectional feedback network and obtaining final classification results.

A. Motivation

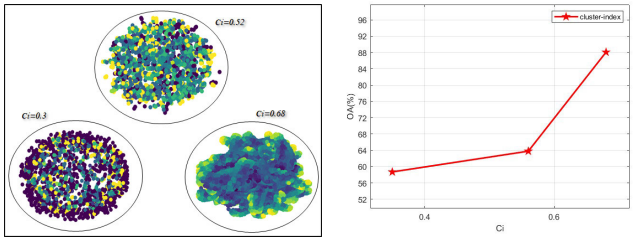
Although SGNN-based methods are able to preserve the structural information between nodes and reduce structural information loss to a certain degree, the intraclass spectral variability and interclass spectral similarity in HSI lead to the imbalance of topological structure within subgraphs. Using the aggregation strategy of fixed K-order neighbors' information with an imbalanced graph structure in the message-passing mechanism makes it difficult to achieve effective smoothing of feature information within subgraphs, leading to over-smoothing or under-smoothing phenomena. In order to illustrate the topological imbalance of each subgraph, we introduce the concept of clustering coefficient [40], which often serves as a metric to describe the cohesion of nodes within a graph structure during network analysis. It measures the degree of connectivity among a node's neighbors, reflecting the topological connectivity of subgraph structure formation among these neighbors. The impact of topological imbalance can be quantified through clustering coefficient analysis, helping to reveal structural differences within subgraphs and their implications for message-passing and feature aggregation. Specifically, for a node i, its clustering coefficient C_i is defined as follows:

$$C_i = \frac{2 * E_i}{d_i * (d_i - 1)} \tag{5}$$

where E_i denotes the number of edges that exist between neighbor nodes of node i, and d_i denotes the degree of node i (that is the number of edges connected to node i). The clustering coefficient ranges from 0 to 1, where it represents the topological connectivity of a subgraph formed between the neighbors of node i.

Next, we conduct related experiments using subgraph structures from the typical Indian Pines dataset due to its imbalanced land-cover distribution and large spectral differences. Fig. 2(a) demonstrates the clustering coefficients of three subgraphs, which are 0.68, 0.52, and 0.3 obtained by the above equation, respectively. The magnitude of the clustering coefficient reflects differences in the topological structure characteristics within a subgraph. In subgraph structures where topological connectivity is dense, the clustering coefficient tends to be larger, while in sparser structures, it is smaller. By examining the value of the clustering coefficient, one can discern variations in topological structure within a subgraph, demonstrating different degrees of subgraph structures. Moreover, Fig. 2(b) presents the overall accuracy (OA) of the SGNN-based method with fixed two convolutional layers on three subgraphs depicted in Fig. 2(a). When applying two-layer convolutions to subgraphs with low clustering coefficients and those with high clustering coefficients, respectively, a significant difference in classification accuracy was observed. These results demonstrate that applying a fixed K-order neighbor information aggregation strategy to subgraphs with different topological connectivity leads to feature information

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(a) (b)

Fig. 2. Indian pines dataset after cutting the molecular map, showing the distribution of each sample point (without edge connections); the color depth represents different categories of samples, C_i represents the clustering coefficient, so as to observe its general topology and spatial structure. (a) Connectivity distribution of three subgraphs. (b) OAs of three subgraphs with two convolution layers.

confusion, ultimately affecting the classification performance. Ideally, adaptive graph convolution aggregation tailored to the inherent connectivity of each subgraph may achieve desirable classification performance in response to the imbalanced connectivity within subgraphs.

B. Subgraph-Dependent Convolution

To address the topological imbalance issue mentioned above, we propose a subgraph-dependent convolution method, which adapts the convolution operation to different subgraph structures, enabling the capture of adaptive-order neighbors and achieving effective feature smoothing within each subgraph. Specifically, we employ the SSAPGCN method to construct the pairwise probability graph structure, as it efficiently integrates spatial and spectral information into the graph structure. The mathematical formulation of this method can be expressed as follows:

$$\min_{\boldsymbol{a}_i^T 1=1, 0\le a_i\le 1} \sum_{i,j=1}^{N} \left(\|\boldsymbol{x}_i-\boldsymbol{x}_j\|_2^2+\delta\|\boldsymbol{l}_i-\boldsymbol{l}_j\|_2^2\right)a_{ij}\Big)+\|\boldsymbol{A}-\boldsymbol{P}\|_F^2 \tag{6}$$

where $\boldsymbol{A}\in R^{N\times N}$ denotes the structure used to capture the pairwise probability connections, $\boldsymbol{P}\in R^{N\times N}$ is the guidance matrix based on expert prior knowledge, $\boldsymbol{a}_i\in R^{1\times N}$ is a vector, a_{ij} denotes the similarity between pixels i and j, and δ denotes the equilibrium hyperparameter between spectral information and spatial correlation. x_i and l_i denote spectral characteristic information and spatial coordinate information respectively.

Next, after obtaining the overall graph structure $\boldsymbol{S}$, we utilize Meit's method for graph segmentation, that is, dividing it into m subgraphs. Consequently, the adjacency matrix A can be expressed as $\boldsymbol{A}=[\boldsymbol{A}_1,\boldsymbol{A}_2,\ldots,\boldsymbol{A}_m]$, while the feature matrix is $\boldsymbol{X}=[\boldsymbol{X}_1,\boldsymbol{X}_2,\ldots,\boldsymbol{X}_m]$. Here, $\boldsymbol{A}_i\in R^{n\times n}$ denotes the adjacency matrix of the ith subgraph, where n denotes the number of nodes of the current subgraph, and $\boldsymbol{X}_i\in R^{n\times d}$ denotes the feature matrix of the ith subgraph, and d denotes the dimension of the current feature. The popular SGNN-based methods usually perform fixed several (i.e., two, three) layers to conduct subgraph convolution. However, it is more reasonable to adopt adaptive subgraph convolution for different subgraphs due to the structural imbalance. Subsequently, for the nodes within the ith subgraph, we determine the kth layer as the optimal convolution layer. Let $\boldsymbol{X}_i^{(0)}=\boldsymbol{X}_i$, where $\boldsymbol{X}_i^{(0)}=[\boldsymbol{x}_1^{(0)},\boldsymbol{x}_2^{(0)},\ldots,\boldsymbol{x}_n^{(0)}]^\top$. Through node-level message aggregation, the node features from the first layer to the $k+1$ layer are expressed as follows:

$$\boldsymbol{x}_i^{(1)}=\boldsymbol{x}_i^{(0)}+\frac{1}{n-1}\sum_{i\ne j}a_{ij}\boldsymbol{x}_j^{(0)}$$
$$\cdots\boldsymbol{x}_i^{(k+1)}=\boldsymbol{x}_i^{(k)}+\frac{1}{n-1}\sum_{i\ne j}a_{ij}\boldsymbol{x}_j^{(k)} \tag{7}$$

where a_{ij} obtained by (12) represents the weight coefficient associated with nodes i and j. Additionally, $\boldsymbol{X}_i^{(k+1)}=[\boldsymbol{x}_1^{(k)},\boldsymbol{x}_2^{(k)},\ldots,\boldsymbol{x}_n^{(k)}]^\top$ denotes the feature matrix from $k+1$th layers neighbor information. Considering that the degree of nodes has a significant impact on the graph's structure and feature propagation, we incorporate the degree information by embedding it as an important feature in the form of residuals. This approach not only better represents the neighboring connections and topological structure of the nodes but also helps alleviate the over-smoothing problem caused by deep training

$$\hat{\boldsymbol{X}}=\beta\boldsymbol{X}+(1-\beta)\boldsymbol{D}_{\boldsymbol{ed}} \tag{8}$$

where $\hat{\boldsymbol{X}}$ represents the residual feature, $\boldsymbol{X}$ represents the input feature, $\boldsymbol{D}_{\boldsymbol{ed}}$ represents the embedded structural degree matrix, and β is the hyperparameter used to balance the input feature and the structural feature matrix.

Furthermore, considering that the determination of the optimal number of layers k significantly influences the quality of neighborhood information aggregation, we design an adaptive adjustment feedback threshold ξ to obtain optimal graph convolution layers based on the varying spectral differences within different subgraphs. To dynamically determine the adaptive threshold value ξ and formulate the iterative optimization method, we adopt clustering performance measure intra(C) [41], [42] to help us to construct adaptive threshold, and the clustering measure leverages the aggregated homogeneous regions to capture inherent spectral variations presented. The adaptive threshold ξ is utilized to establish the similarity threshold within and between classes. The adjustment of the number of k iterations (corresponding kth layers) can be guided by the clustering metric. The specific formula can be defined as follows:

$$\xi=\text{intra}(C)=\frac{1}{|C|}\sum_{i=1}^{|C|}\frac{1}{|c_i|^2}\sum_{\boldsymbol{x}_i,\boldsymbol{x}_j\in c_i}\|\hat{\boldsymbol{x}}_i-\hat{\boldsymbol{x}}_j\|_2^2 \tag{9}$$

where C denotes the category, $|C|$ denotes the number of categories, c_i denotes the class i, and $|c_i|$ denotes the number of samples in class i. The metric intra(C) denotes the degree of similarity among samples within each class. The reasons for using intraclass distance as a measurement standard are as follows. Subgraph convolution is essentially a form of Laplacian smoothing [43], which tends to make the features between adjacent samples more similar, thus reducing the distance between samples. The spectral features within the subgraph become smoother as k increases, which could significantly reduce the intracluster distance. Intuitively, imbalanced

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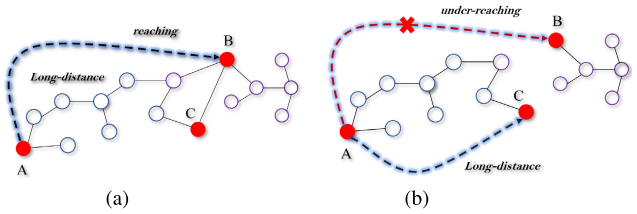


Fig. 3. Changes in sample remote dependence before and after subgraph segmentation, and whether it can be used to display aggregated information. The black dotted line represents reachability, while the red dotted line represents unreachability. (a) Long-range structural dependency of the whole graph. (b) Long-range structural dependency between subgraphs.

subgraph structures require different numbers of subgraph convolution layers to achieve effective smoothing. The intra(C) can effectively measure the degree of feature smoothing within a subgraph. The subgraph with sparse connectivity may require more convolution layers, while the subgraph with strong connectivity may require fewer convolution layers to achieve effective feature smoothing.

Here, we set $\xi = \text{intra}(C)$ to be used as the early stopping condition for adaptive subgraph iteration. The adaptive iteration strategy is as follows. For each epoch, we calculate the intra(C) for the current round, and set the intraclass distance between round t and round $t+1$ as $\xi^{(t)}$ and $\xi^{(t+1)}$, respectively. If $\xi^{(t)} < \xi^{(t+1)}$, we consider the features to be reasonably similar and proceed to the next iteration. If $\xi^{(t)} > \xi^{(t+1)}$, we stop the iteration process at this point and record the current iteration layer t. Stopping the iteration when the first minimum occurs can prevent the occurrence of the over-smoothing phenomenon and ensure the effectiveness of the training process.

C. Subgraph Optimization

Considering that the quality of subgraphs is susceptible to the influence of spectral variability in HSI, and that the subgraphs predominantly take into account only local characteristics, disrupting global dependencies among nodes within a subgraph, we develop the strategy of long-range dependency. The basic idea of long-range dependency within a subgraph is illustrated in Fig. 3. Taking nodes A and B as an example in Fig. 3(a), they belong to the same class via multihop neighbors due to its intraclass spectral feature variation. For node A, it would be beneficial to achieve the effective intraclass feature smoothing when aggregating neighboring information from node B due to its feature diversity. However, in practical applications, although nodes A and B are connected in the complete large graph, it is entirely possible for graph partitioning algorithms to sever the connectivity between nodes A and B, dividing them into two separate subgraphs. Constrained by the training strategy of SGNNs, it is not feasible to capture the information of node B across subgraphs. Therefore, we propose a strategy that captures the neighbor node C of node B within the subgraph, thereby achieving the goal of capturing long-range dependency information within the subgraph. The basic idea behind this approach is that since nodes B and C are connected within the entire large graph and are each other's first-order neighbor, it indicates that nodes B and C share strong semantic feature similarity. When node C also happens to be within the subgraph containing node A, we treat node C as a first-order neighbor of node A for effective feature aggregation in the message-passing paradigm during the training phase of SGDNet. The specific graphical representation is shown in Fig. 3(b).

Next, when considering optimizing the subgraph structure from the structurally long-range distance aspect, the topological positions of labeled nodes in subgraphs are the concern. Structural distances significantly influence the effectiveness of labeled information under the framework of the message-passing mechanism within subgraphs. Hence, optimizing the subgraph structure from a distance perspective is necessary to establish the compact intraclass relationships between labeled nodes and unlabeled nodes [44]. We divide the set of labeled nodes $\mathcal{V}_L$ into subsets denoted as anchor points $\{\boldsymbol{V}_L^1, \boldsymbol{V}_L^2, \ldots, \boldsymbol{V}_L^C\}$. Each subset $\boldsymbol{V}_L^c$ represents labeled nodes of a specific class. This class-based set of anchor points is designed to differentiate between various class-related information. Rather than treating all labeled nodes uniformly, this approach acknowledges class distinctions, enhancing the training process. Concretely, for any node, we introduce a function $\phi(\cdot,\cdot)$ to gauge its positional relationship with respect to the anchor set within a graph. This function can be defined based on the connectivity among the graph's nodes

$$\boldsymbol{p}_i = \left(\phi\left(\boldsymbol{v}_i, \boldsymbol{V}_L^1\right), \phi\left(\boldsymbol{v}_i, \boldsymbol{V}_L^2\right), \ldots, \phi\left(\boldsymbol{v}_i, \boldsymbol{V}_L^C\right)\right) \tag{10}$$

where $\phi(\boldsymbol{v}_i, \boldsymbol{V}_L^c)$ acts as the position coding function, hinging on the connectivity between node $\boldsymbol{v}_i$ and the anchor set $\boldsymbol{V}_L^c$. The definition of $\phi(\boldsymbol{v}_i, \boldsymbol{V}_L^c)$ is the average length of the shortest path between node v_i and the nodes in anchor set $\boldsymbol{V}_L$, indicating the strength of their connection

$$\phi\left(\boldsymbol{v}_i, \boldsymbol{V}_L^c\right) = \frac{\sum_{\boldsymbol{v}_j \in \mathcal{N}_c(v_i)} |\boldsymbol{P}_{\boldsymbol{sp}}\left(\boldsymbol{v}_i, \boldsymbol{v}_j\right)|}{|\mathcal{N}_c(v_i)|} \tag{11}$$

where $\mathcal{N}_c(v_i)$ denotes the nodes connected with $\boldsymbol{v}_i$ in the subset $\boldsymbol{V}_L^c$, and $\boldsymbol{P}_{\boldsymbol{sp}}(\boldsymbol{v}_i, \boldsymbol{v}_j)$ signifies the length of the shortest path between nodes $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$. Subsequently, we proceed to transform the positional encoding into a $\boldsymbol{d}_0$ dimensional space

$$\boldsymbol{h}_i^{\boldsymbol{p}} = \boldsymbol{W}_\phi \cdot \boldsymbol{p}_i. \tag{12}$$

Here, $\boldsymbol{W}_\phi$ denotes a trainable vector. The purpose of this vector is to ensure that if two nodes possess similar shortest paths to the anchor set, their position codings will be similar.

After acquiring the position coding, we utilize a measurement function that takes into account both node feature information and location similarity to estimate the probability of edge existence. In this particular case, we have chosen the widely used multihead cosine similarity function

$$a_{ij}^P = \frac{1}{m}\sum_{h=1}^{m} \cos\left(\boldsymbol{W}_{\boldsymbol{h}} \cdot \left(\boldsymbol{z}_i || \boldsymbol{h}_i^{\boldsymbol{p}}\right), \boldsymbol{W}_{\boldsymbol{h}} \cdot \left(\boldsymbol{z}_j || \boldsymbol{h}_j^{\boldsymbol{p}}\right)\right) \tag{13}$$

where m denotes the number of heads, $\boldsymbol{W}_{\boldsymbol{h}}$ denotes the weight matrix of the hth head, $\boldsymbol{z}_{\boldsymbol{i}}$ denotes the representation vector of node $\boldsymbol{v}_{\boldsymbol{i}}$, and $||$ signifies concatenation. After obtaining the structural position associations of each node with all labeled nodes, we concatenate the structural position encoding with its own features. The purpose of this approach is to enhance the

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homogeneous connectivity relationships between each node and nodes within the same class, and also to effectively separate the dissimilar relationships between samples of different classes. The reason for using cosine similarity to measure the similarity between nodes after feature concatenation is that it controls the similarity and dissimilarity between nodes within the range of -1 to 1, rather than confining the differences between them from 0 to 1. This approach better distinguishes the gap between homogeneous and heterogeneous samples, facilitating subsequent sparse processing of structural similarities. From this analysis, we obtain the magnitude of similarity between nodes i and j, allowing us to construct the complete similarity matrix $\boldsymbol{A}$. Given that a_{ij}^P typically falls within the range of $[-1,1]$, it is important to ensure that the resulting adjacency matrix remains non-negative to preserve effective edges between any two nodes. Moreover, in order to reduce computational complexity and remove unnecessary noisy connections, we could derive a symmetric and sparse adjacency matrix $\boldsymbol{A}'$ from $\boldsymbol{A}$ by considering only the ε-neighborhood of each node. In this process, we mask out those elements of $\boldsymbol{A}$ that are below a predefined non-negative threshold ε

$$\boldsymbol{A}' = \begin{cases} a_{ij}^P, & a_{ij}^P > \varepsilon \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

As mentioned earlier, we employ a multihead cosine similarity function to take into account both node feature information and positional similarity. This function allows us to estimate the likelihood of edge existence, subsequently enabling the reconstruction of the graph's structure and the formulation of edge expressions, which in turn facilitates subgraph optimization. Subsequently, we align the optimized graph structure with the pre-optimized graph structure. It is our belief that the pre-optimized graph structure encapsulates valuable spatial and spectral information, which aims to judiciously preserve

$$\boldsymbol{A}_{\boldsymbol{g}} = \boldsymbol{A} + \alpha \boldsymbol{A}' \tag{15}$$

where $\boldsymbol{A}_{\boldsymbol{g}}$ denotes the adjacency matrix of the optimized graph as the GCN input, $\boldsymbol{A}$ denotes the adjacency matrix of the optimized pre-graph, and α is the tradeoff coefficient between the adjacency matrix of the optimized pre-graph and the similarity matrix $\boldsymbol{A}'$.

In the training stage, we propose to introduce an additional regularization loss (RL_loss) term based on the previous cross-entropy loss function. This regularization term is introduced by aligning the optimized adjacency matrix with the data, aiming to enhance the overall model's generalization capabilities. From the perspective of subgraph optimization, the regularization loss is beneficial to smooth node features. In other words, this regularization ensures that node features exhibit certain similarity, which is beneficial for capturing structural similarity within the graph and mitigating topology imbalances in subgraph structures. By obtaining and leveraging these feature similarities, we can achieve more effective feature smoothing and create a positive feedback loop, resulting in a more informative representation

$$f(A, X) = \mathcal{L} + \frac{1}{2n^2} \boldsymbol{A}_{ij}^{\boldsymbol{g}} \left\| \hat{\boldsymbol{x}}_i - \hat{\boldsymbol{x}}_j \right\|^2 + \frac{\gamma}{n^2} \|\boldsymbol{A}_{\boldsymbol{g}}\|_F^2 \tag{16}$$

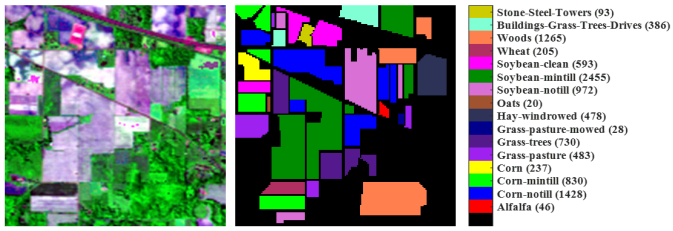
where $\hat{\boldsymbol{x}}_i$ denotes the features of node i, $\boldsymbol{A}_{ij}^{\boldsymbol{g}}$ and $\boldsymbol{A}_{\boldsymbol{g}}$ represent the adjacency matrix of the optimized subgraph, and γ is a non-negative hyperparameter that controls the sparsity of the graph. The function $f(\boldsymbol{A}, \boldsymbol{X})$ denotes the overall loss, comprising the cross-entropy loss of the subgraph, denoted as $\mathcal{L}$ in the first term. The second term acts as a regularization loss, promoting similarity in features across different sample points through the use of gradient descent. The third term is introduced to prevent the occurrence of the trivial solution $\boldsymbol{A}_{\boldsymbol{g}} = 0$. The entire process of SGDNet is outlined in Algorithm 1.

Algorithm 1 SGDNet Method

Input: Feature matrix $\mathbf{X}$ and the initial graph $\mathbf{G}$ from datasets.
Output: Labels of test samples.
1) Obtain the adjacency matrix $\mathbf{A_g}$ through subgraph optimization, and calculate the residual feature $\hat{\mathbf{X}}$ through the embedding degree matrix and feature matrix.
2) Obtain the aggregation matrix $\mathbf{X}^{(0)} \leftarrow GCN(\mathbf{A}_{\mathbf{g}}^{(0)}, \mathbf{X}^{(0)})$
3) Obtain the feature matrix through the aggregation matrix and the residual matrix $\mathbf{X}^{(0)} \leftarrow \mathbf{X}^{(0)} + \hat{\mathbf{X}}^{(0)}$
4) Obtain the intraclass distance $\xi \leftarrow Eq.(8)$
5) Obtain the loss $\mathcal{L}^{(0)} \leftarrow f(\mathbf{A}_{\mathbf{g}}^{(0)}, \mathbf{X}_{\mathbf{cur}}^{(0)})$
6) **While** (intra$(C) \geq \xi$)
7) $k \leftarrow k + 1$
8) Obtain the similarity matrix by using multi-head cosine similarity function and sparse procesing $\mathbf{A}' \leftarrow$ Eq. (12)-(13).
9) Calculate updated adjacency matrix $\mathbf{A}_{\mathbf{g}}^{(k)} \leftarrow$ Eq. (14).
10) Obtain the aggregation matrix $\mathbf{X}^{(k)} \leftarrow GCN(\mathbf{A}_{\mathbf{g}}^{(k)}, \mathbf{X}^{(k)})$
11) Obtain the intraclass distance $\xi \leftarrow Eq.(8)$
12) Obtain the loss $\mathcal{L}^{(k)} \leftarrow f(\mathbf{A}_{\mathbf{g}}^{(k)}, \mathbf{X}_{\mathbf{cur}}^{(k)})$
13) **End While**
14) Obtain the total loss $\mathcal{L} \leftarrow \mathcal{L}^{(0)} + \sum_{i=1}^{k} \mathcal{L}^{(i)}/k$

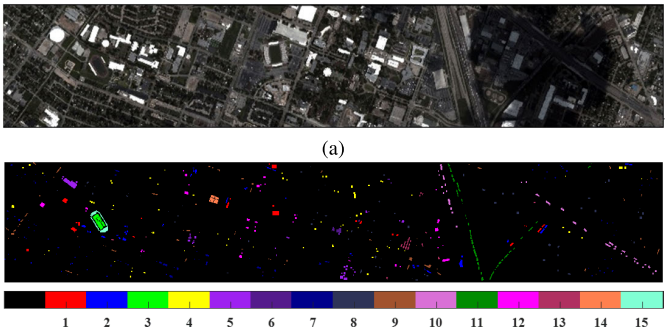
Finally, after the subgraph optimization at the distance and feature levels, we introduce the key steps of adaptive convolution. In this step, two critical underlying factors are considered. First, the optimized subgraph structure enables effective aggregation of more relevant nodes. By aggregating nodes with similar features and semantic relationships, adaptive convolution can more accurately capture the relationships between nodes, further improving model performance. Second, the fusion of subgraph optimization and adaptive convolution forms an optimization-convolution network, constructing a forward feedback network structure. This forward feedback mechanism strengthens the model's learning ability on specific task data. By introducing adaptive convolution based on subgraph optimization, the model can better understand the local and global features of the data. Simultaneously, through the feedback of adaptive convolution, the subgraph structure is continuously optimized. This dual optimization forward feedback loop enhances the model's expressive power and learning ability, improving the modeling effectiveness of complex relationships. Therefore, by introducing adaptive

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(a) (b)

Fig. 4. Indian Pines data. (a) False color image (bands 20, 69, and 178 are red, green, and blue, respectively). (b) Ground-truth image.



(a)

(b)

Fig. 5. Houston data. (a) False color image (bands 45, 38, and 32 are red, green, and blue, respectively). (b) Ground-truth image.

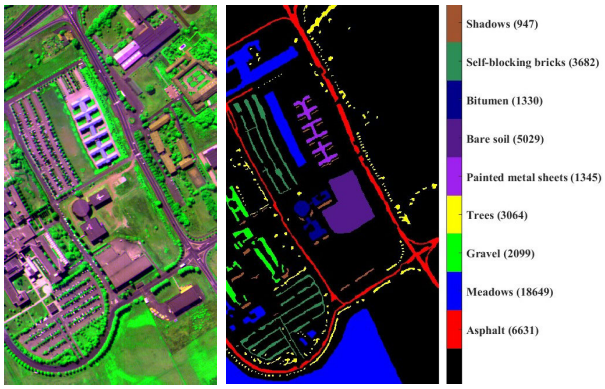
convolution, we construct a more targeted and adaptive network structure, laying the foundation for efficient information propagation.

IV. Experimental Results and Analysis

In this section, we evaluate the performance of the proposed algorithm using three widely used HSI datasets. We compare the algorithm with other state-of-the-art algorithms, and the results demonstrate its superiority in terms of classification performance.

A. Dataset Description

1) The Indian Pine dataset was acquired using the AVIRIS sensor, which captures detailed information across 220 spectral bands. The image size is 145 × 145 pixels, with a spatial resolution of 20 m and a spectral resolution of 10 nm. The scene comprises 16 land-cover classes, representing a diverse agricultural field. Fig. 4(a) and (b), respectively, shows false-color images and the ground truth of the Indian Pine dataset.
2) The PaviaU dataset was acquired using the ROSIS sensor, capturing spectral information across 103 bands. It is a large dataset with a size of 610 × 340 pixels, essential for land-use and land-cover analysis. Fig. 5(a) shows the false-color representation for intuitive visualization of the spectral information, while Fig. 5(b) contains the corresponding ground-truth image for classification evaluation.
3) The Houston University (Houston) dataset was acquired using the CASI sensor, containing 349 × 1905 pixels with 144 spectral bands. The image was captured over the University of Houston campus and the neighboring urban area in June 2013. With a spatial resolution of 2.5 m, this dataset is valuable for land-use and land-cover analysis. Fig. 6(a) and (b) shows the false-color representation and corresponding ground truth, respectively, facilitating classification evaluation.



(a) (b)

Fig. 6. PaviaU data. (a) False color image (bands 20, 69, and 178 are red, green, and blue, respectively). (b) Ground-truth image.

B. Experimental Settings

To assess the superiority of the proposed SGDNet model, we conducted experiments comparing its performance in HSI classification with several existing models, including ASSMN [45], 3DCNN [46], DSGSF [47], morphFormor [21], GraphGST [22] DPRN [48], GCN [49], Cluster-GCN [32], MiniGCN [25], and FuNet-C [25]. In our experiments, to reduce computational complexity, we applied principal component analysis (PCA) to the three 60-D HSI datasets. Subsequently, for the three datasets including Indian Pines, Houston, and PaviaU, we randomly selected 1.5%, 1%, and 0.5% of labeled samples from each category for training, respectively, and assigned the remaining samples for testing, unless otherwise indicated. The table presents the number of training and test samples used in the experiments (see Table I). We conducted each method ten times to mitigate bias stemming from random sampling. Finally, we employed the OA, Kappa coefficient (KC), and average accuracy (AA) to evaluate the performance of all methods in HSI classification [50]. Higher values for these metrics indicate better classification results.

All experiments were conducted on a Linux server equipped with an Intel Core i7-9700K CPU, NVIDIA GeForce RTX 3090 GPUs, and 24 GB of RAM. The operating system used was Ubuntu 20.04 LTS. Moreover, the experiments were carried out in a Python 3.10 environment, with the following libraries and their respective versions: PyTorch 2.0.1+cu11.7, NumPy 1.24.1, SciPy 1.10.1, and Metis 0.2a.4. The used hyperparameters in experiments were as follows. The learning rate was set to 0.03, the hyperparameter α was set to 0.2, the hyperparameter β was set to 0.2, and the dropout rate was set to 0.01 in order to prevent overfitting.

C. Experimental Results

1) The experimental results on the Indian Pines dataset are shown in Fig. 7 and Table II, demonstrating that

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TABLE I

NUMBER OF TRAINING AND TESTING SAMPLES ON THE THREE DATASETS

Indian Pines				PaviaU				Houston			
ID	Name	train	test	ID	Name	train	test	ID	Name	train	test
1	Alfalfa	1	45	1	Asphalt	38	6593	1	Healthy grass	15	1236
2	Corn-notill	24	1404	2	Meadows	98	18551	2	Stressed grass	16	1238
3	Corn-mintill	15	815	3	Gravel	15	2084	3	Synthetic grass	10	687
4	Corn	5	232	4	Trees	21	3043	4	Tree	16	1228
5	Grass-pasture	10	473	5	Painted metal sheets	9	1336	5	Soil	15	1227
6	Grass-trees	14	716	6	Bare Soil	30	4999	6	Water	6	319
7	Grass-pasture-mowed	1	27	7	Bitumen	12	1318	7	Residential	15	1253
8	Hay-windrowed	8	470	8	Self-Blocking Bricks	23	3659	8	Commercial	15	1229
9	Oats	2	18	9	Shadows	9	938	9	Road	16	1236
10	Soybean-notill	18	954		Total	255	42521	10	Highway	16	1211
11	Soybean-mintill	40	2415					11	Railway	15	1220
12	Soybean-clean	12	581					12	Parking lot 1	16	1217
13	Wheat	7	198					13	Parking lot 2	8	461
14	Woods	23	1242					14	tennis court	7	421
15	Buildings-Grass-Trees-Drives	9	377					15	Running track	10	650
16	Ston-Steel-Towers	4	89						Total	196	14833
	Total	193	10056								

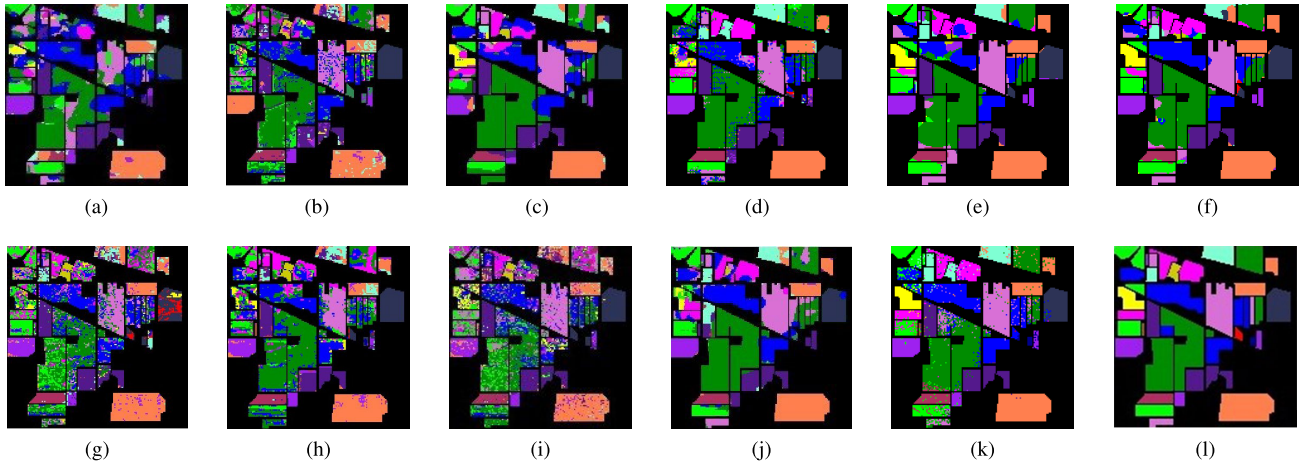


Fig. 7. Classification maps of different methods on the Indian Pines dataset. (a) 3DCNN. (b) ASSMN. (c) DPRN. (d) DSGSF. (e) morphFormer. (f) GraphGST. (g) Cluster-GCN. (h) Funet-C. (i) MiniGCN. (j) GCN. (k) SGDNet. (l) Ground truth.

TABLE II

OA (STD) (%), AA (STD) (%), KC (STD) (%) OF DIFFERENT COMPARED METHODS AND SGDNet METHOD FOR INDIAN DATASET

Class	3DCNN	DPRN	ASSMN	DSGSF	morphFormer	GraphGST	GCN	Funet-C	Cluster-GCN	MiniGCN	SGDNet
1	8.65(0.30)	19.46(9.85)	74.44(26.49)	58.01(33.71)	**99.80(0.70)**	99.71(0.63)	60.70(33.90)	14.44(6.39)	11.11(0.00)	25.78(24.48)	98.41(2.16)
2	62.82(0.29)	61.38(3.13)	67.26(8.18)	74.11(8.77)	79.70(8.89)	**87.43(3.38)**	77.14(8.97)	57.13(5.56)	61.52(1.56)	51.34(7.39)	82.16(4.17)
3	45.67(0.28)	67.38(4.55)	69.48(9.20)	72.83(9.94)	87.50(6.16)	**87.71(7.31)**	77.05(5.41)	42.42(6.25)	42.53(4.27)	44.29(5.02)	81.50(4.40)
4	38.34(0.32)	37.09(14.25)	64.91(11.18)	64.50(14.65)	86.90(14.00)	55.58(26.59)	86.91(8.12)	23.32(6.95)	52.12(0.00)	21.94(7.68)	**92.37(6.46)**
5	81.78(0.27)	87.89(3.54)	65.20(10.43)	82.48(8.65)	**87.90(5.39)**	85.65(5.06)	81.68(6.93)	61.40(13.59)	64.00(0.00)	72.62(12.57)	87.72(3.52)
6	89.33(0.22)	92.94(5.34)	90.59(4.44)	88.34(5.87)	96.60(3.82)	**98.01(2.61)**	85.09(7.93)	93.42(5.78)	65.70(1.65)	93.83(4.18)	97.79(1.09)
7	36.45(0.30)	54.55(23.39)	83.33(13.89)	**85.06(18.68)**	97.50(3.23)	98.62(1.93)	80.00(36.90)	38.51(23.95)	42.50(0.00)	51.11(26.12)	64.44(36.42)
8	92.83(0.25)	98.81(1.22)	94.64(9.15)	91.09(4.47)	99.70(0.95)	**100.00(0.00)**	96.30(7.55)	97.38(2.74)	88.91(3.32)	79.85(12.14)	99.96(0.09)
9	81.85(0.30)	38.64(12.38)	83.88(11.25)	71.81(35.72)	90.00(31.60)	25.79(41.99)	54.21(29.67)	65.56(21.08)	3.53(0.00)	48.89(18.29)	**91.67(7.97)**
10	52.05(0.30)	65.56(6.11)	64.50(18.54)	73.44(9.09)	84.70(8.93)	87.94(3.75)	81.35(4.86)	57.31(4.71)	46.03(0.00)	52.62(9.65)	**88.40(3.71)**
11	67.91(0.25)	78.59(4.21)	87.52(5.75)	83.47(9.68)	74.50(5.65)	88.18(2.94)	87.82(5.21)	77.32(4.46)	78.78(0.00)	53.24(4.45)	**88.43(1.97)**
12	44.77(0.28)	45.37(10.23)	65.31(8.12)	61.10(10.77)	**80.20(17.10)**	69.49(14.76)	75.55(6.93)	43.14(8.85)	63.49(3.62)	40.07(6.22)	79.74(6.57)
13	93.00(0.18)	96.83(2.76)	97.12(3.21)	87.32(6.00)	**99.40(0.33)**	99.21(0.42)	85.32(15.93)	97.92(2.97)	38.94(0.00)	97.27(1.67)	96.87(2.83)
14	93.25(0.16)	91.78(2.76)	95.04(2.51)	90.41(3.32)	94.70(7.26)	**98.66(2.03)**	92.51(7.01)	89.91(4.89)	87.57(0.62)	82.25(5.12)	95.43(3.37)
15	45.62(0.27)	60.99(4.49)	75.88(6.95)	75.70(15.49)	**95.80(5.21)**	74.22(13.62)	78.63(9.50)	31.86(9.31)	60.58(0.00)	24.14(7.24)	87.72(7.88)
16	14.25(0.30)	81.44(11.32)	94.61(5.47)	83.21(12.63)	**99.80(0.52)**	58.91(50.79)	57.03(32.25)	91.34(5.53)	27.36(0.00)	87.08(1.32)	90.79(13.94)
OA	67.85(0.30)	74.23(2.87)	79.41(2.11)	79.79(5.54)	85.00(1.80)	87.56(1.27)	83.73(2.36)	67.87(1.31)	66.32(0.53)	59.28(1.48)	**88.76(0.48)**
AA	59.29 (0.07)	67.33(3.56)	79.61(2.98)	77.68(8.42)	**91.00(3.13)**	74.78(6.47)	81.47(2.66)	57.78(2.18)	52.17(1.48)	54.49(2.06)	88.96(1.82)
KC	62.71 (0.26)	70.58(3.27)	76.40(2.49)	76.86(6.43)	83.50(2.00)	85.79(1.49)	78.58(3.74)	63.00(1.43)	61.64(0.73)	53.86(1.60)	**87.21(0.55)**

SGDNet achieves an OA exceeding 85%, significantly outperforming other popular methods and achieving outstanding performance in complex land-cover types. CNN-based methods such as 3DCNN, DPRN, ASSMN, and DSGSF are limited by fixed receptive fields, where they struggle to capture the diversity of local details and

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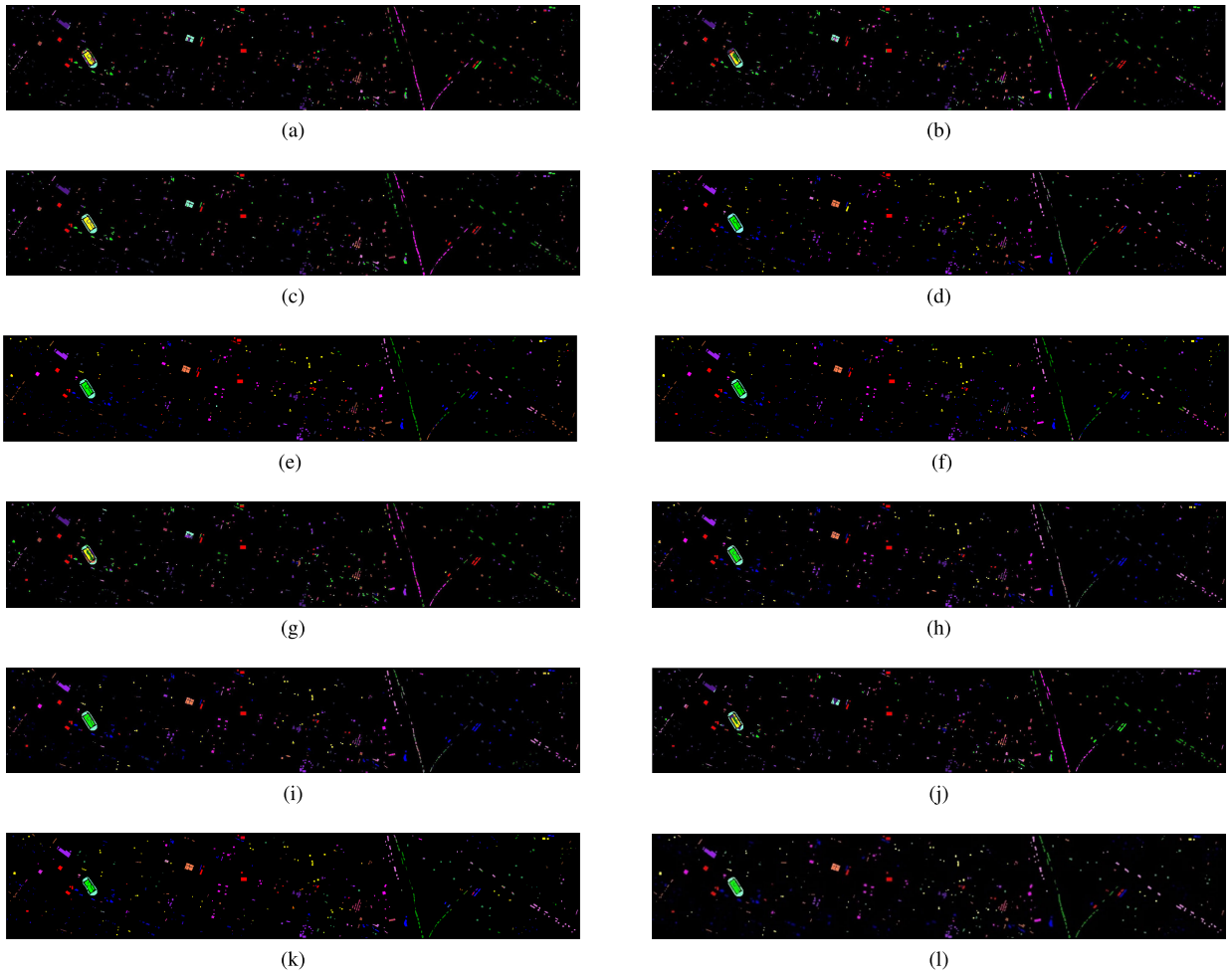


Fig. 8. Classification maps of different methods on the Houston dataset. (a) 3DCNN. (b) ASSMN. (c) DPRN. (d) DSGSF. (e) morphFormer. (f) GraphGST. (g) Cluster-GCN. (h) Funet-C. (i) MiniGCN. (j) GCN. (k) SGDNet. (l) Ground truth.

achieve OA values of 67.85%, 74.23%, 79.41%, and 79.79%, respectively. Meanwhile, GCN-based methods such as morphFormer and GraphGST are able to achieve flexible information aggregation, achieving good OA values of 85.00% and 87.56%, respectively. However, due to their fixed hop-based neighborhood aggregation mechanism, they are prone to either under-smoothing or over-smoothing. SGDNet method addresses these issues by dynamically adjusting the hops of aggregating neighborhood information, optimizing the structural expression, and effectively capturing long-range dependencies through an adaptive aggregation mechanism, achieving more precise classification in complex scenarios. As illustrated in Fig. 7, the classification results of SGDNet closely align with the true land-cover distribution, exhibiting clear boundaries.

2) The experimental results on the Houston dataset are shown in Table III and Fig. 8, indicating that SGDNet achieves optimal results in both OA and KC values. The Houston dataset possesses high spatial resolution, diverse land-cover types, and complex boundaries, which pose great challenges for the classification model in terms of capturing local details and modeling rich context information. SGDNet enhances its ability to distinguish heterogeneous land covers by leveraging the intrinsic structural features of the subgraph, particularly excelling in classifying urban areas, such as roads, buildings, and vegetation. Its adaptive aggregation strategy prevents the redundancy of local information caused by fixed neighborhood ranges in traditional GCN-based methods, while the dynamic adjustment of subgraph structures alleviates both under-smoothing and over-smoothing problems. The experimental results demonstrate that SGDNet not only leads in OA but also excels in the fidelity of the classification map, offering a more accurate reflection of the true land-cover distribution in Fig. 8.

3) The experimental results on the PaviaU dataset are shown in Table IV and Fig. 9, further validating the superiority of the SGDNet method. The PaviaU dataset primarily consists of urban areas, containing many land-cover types with complex textures, such as asphalt pavement, bricks, and shadowed regions, making the classification task more challenging. SGDNet achieves optimal results in OA, AA, and KC. As shown in the classification map, SGDNet exhibits better classification performance, achieving an OA of 93.17%, AA of 91.15%, and KC of 90.97%, compared to other

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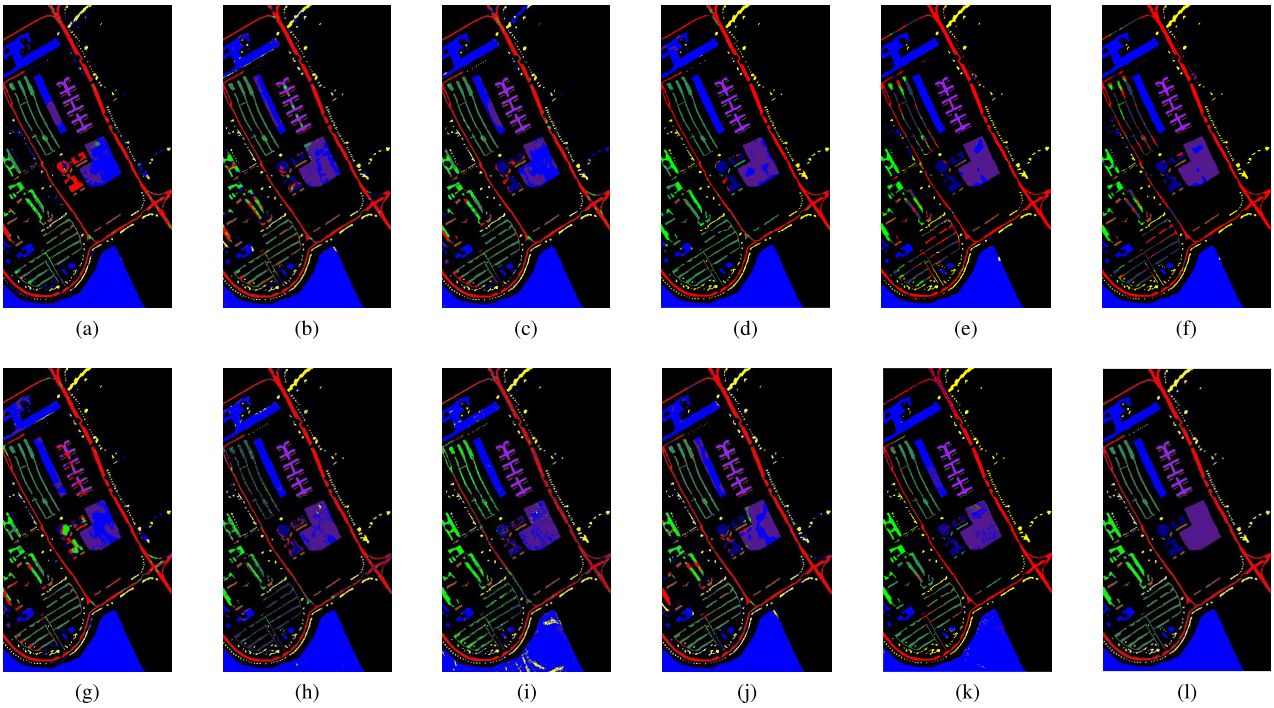


Fig. 9. Classification maps of different methods on the PaviaU dataset. (a) 3DCNN. (b) ASSMN. (c) DPRN. (d) DSGSF. (e) morphFormer. (f) GraphGST. (g) Cluster-GCN. (h) Funet-C. (i) MiniGCN. (j) GCN. (k) SGDNet. (l) Ground truth.

TABLE III

OA (STD) (%), AA (STD) (%), KC (STD) (%) OF DIFFERENT COMPARED METHODS AND SGDNet METHOD FOR HOUSTON DATASET

Class	3DCNN	DPRN	ASSMN	DSGSF	morphFormer	GraphGST	GCN	Funet-C	Cluster-GCN	MiniGCN	SGDNet
1	87.39(0.01)	90.95(8.12)	87.30(3.69)	86.98(7.33)	**91.64(6.04)**	91.54(6.23)	70.76(10.93)	84.67(26.57)	63.19(25.40)	97.98(4.41)	84.48(4.63)
2	76.08(0.02)	**97.81(2.18)**	52.14(7.16)	85.35(6.15)	89.91(5.96)	91.64(6.73)	58.52(13.46)	85.00(26.73)	52.88(16.48)	90.15(4.79)	85.67(6.80)
3	87.60(0.01)	97.33(2.21)	76.87(11.44)	99.33(1.30)	86.88(11.55)	83.99(24.79)	**99.80(0.39)**	85.99(26.94)	99.77(0.17)	99.01(0.32)	98.96(2.86)
4	85.64(0.01)	**90.89(2.08)**	64.43(8.01)	88.04(2.58)	90.27(5.31)	90.61(2.94)	42.41(6.41)	83.26(26.52)	54.74(19.06)	90.77(2.66)	85.28(4.0)
5	87.52(0.01)	92.57(4.61)	89.35(6.71)	87.55(3.65)	97.76(5.62)	97.34(7.96)	90.20(7.79)	87.64(27.46)	81.32(14.45)	97.74(2.09)	**98.29(0.70)**
6	14.84(0.27)	50.18(35.83)	57.68(10.19)	**97.76(2.29)**	60.78(38.45)	69.57(32.00)	88.79(7.09)	75.99(24.09)	76.59(17.09)	94.08(6.01)	88.70(5.99)
7	50.60(0.01)	70.13(6.81)	64.78(11.27)	72.18(7.27)	67.74(12.20)	68.91(9.80)	55.27(6.19)	64.90(20.36)	57.62(15.80)	55.13(7.64)	**84.66(6.75)**
8	54.51(0.02)	68.92(9.26)	50.37(8.38)	**70.99(13.01)**	63.73(7.36)	64.74(7.97)	40.54(6.90)	59.82(19.63)	39.47(15.09)	68.05(11.46)	59.01(5.93)
9	51.90(0.01)	66.55(10.94)	50.31(8.15)	57.87(14.95)	62.48(12.83)	**75.24(10.01)**	50.09(5.83)	63.26(20.34)	30.59(14.16)	39.45(18.10)	63.44(8.89)
10	54.40(0.01)	82.49(11.12)	71.83(6.14)	56.36(10.57)	65.34(21.69)	73.68(12.51)	76.78(10.37)	65.98(21.30)	47.83(24.79)	68.54(14.42)	**91.89(4.72)**
11	57.81(0.01)	74.81(9.71)	62.59(14.34)	57.36(17.70)	76.30(12.23)	76.32(12.56)	77.37(9.91)	56.50(17.71)	58.44(34.08)	66.53(11.61)	**86.25(6.53)**
12	50.46(0.02)	76.55(7.04)	62.45(10.93)	54.93(15.38)	61.80(25.30)	73.26(17.76)	67.16(8.25)	69.73(21.84)	45.94(16.27)	**79.52(13.34)**	77.07(6.43)
13	27.33(0.03)	60.10(9.86)	42.08(8.61)	67.60(19.08)	61.25(23.48)	61.80(25.30)	61.87(16.95)	43.61(14.44)	41.27(23.56)	24.34(9.82)	**74.19(7.73)**
14	79.70(0.02)	91.11(5.68)	72.18(14.17)	94.03(4.74)	98.40(1.77)	91.90(18.09)	87.07(17.07)	74.33(23.27)	69.88(32.78)	91.30(10.18)	**99.38(1.95)**
15	93.27(0.02)	97.79(3.01)	75.86(7.10)	94.47(6.24)	97.99(3.87)	**100.00(0.00)**	99.17(0.29)	88.30(27.76)	79.77(40.62)	99.06(0.38)	97.94(2.61)
OA	67.40(0.01)	81.09(2.24)	65.80(3.54)	74.49(6.58)	78.71(1.84)	80.85(2.34)	67.38(3.07)	72.52(22.71)	57.11(6.12)	76.76(2.91)	**83.54(1.17)**
AA	63.94(0.02)	80.26(2.11)	65.35(3.69)	78.05(5.60)	78.84(3.00)	80.67(2.43)	64.74(3.32)	67.87(21.25)	53.63(6.19)	72.60(2.53)	**85.01(1.14)**
KC	63.38(0.02)	79.93(2.55)	63.04(3.82)	72.39(7.13)	76.98(1.99)	79.29(2.52)	71.05(3.04)	71.02(22.28)	59.96(7.20)	74.88(3.13)	**82.21(1.26)**

TABLE IV

OA (STD) (%), AA (STD) (%), KC (STD) (%) OF DIFFERENT COMPARED METHODS AND SGDNet METHOD FOR PaviaU DATASET

Class	3DCNN	DPRN	ASSMN	DSGSF	morphFormer	GraphGST	GCN	Funet-C	Cluster-GCN	MiniGCN	SGDNet
1	85.49(0.71)	91.51(2.63)	94.41(3.17)	91.25(3.30)	**95.23(1.78)**	95.77(1.65)	80.07(15.01)	83.67(6.70)	66.71(2.38)	74.81(5.62)	90.07(2.21)
2	86.44(0.73)	97.01(1.28)	**98.47(1.69)**	95.82(2.41)	96.95(1.64)	97.16(1.54)	96.77(3.44)	95.05(1.33)	95.56(1.61)	93.90(1.62)	96.39(2.24)
3	28.26(2.66)	37.96(9.38)	61.60(9.59)	80.32(10.63)	85.22(9.77)	84.80(4.90)	70.32(9.75)	70.13(11.84)	83.18(7.05)	85.95(5.52)	**87.72(8.02)**
4	88.06(0.41)	84.35(2.95)	92.25(4.02)	**98.71(1.27)**	83.59(6.97)	91.46(2.27)	83.82(4.14)	87.10(4.15)	22.24(0.00)	87.81(3.76)	89.55(13.39)
5	96.77(0.56)	97.99(2.06)	90.04(9.22)	98.36(1.60)	**99.97(0.04)**	99.95(0.04)	99.88(0.04)	99.25(0.28)	96.27(0.00)	99.29(0.23)	99.87(0.20)
6	45.00(2.86)	65.92(6.34)	88.24(6.39)	90.82(9.30)	63.73(12.03)	66.46(7.38)	90.06(6.99)	69.47(5.61)	**98.62(0.00)**	67.31(3.61)	91.06(4.80)
7	70.68(1.19)	67.38(6.82)	78.07(12.78)	78.35(7.35)	75.65(14.83)	85.50(27.02)	12.96(21.91)	74.67(9.36)	79.95(8.03)	83.41(11.09)	**97.26(2.00)**
8	56.68(2.83)	79.40(6.31)	87.97(3.94)	81.32(5.52)	87.80(8.09)	92.45(3.42)	**94.71(2.76)**	66.72(9.29)	74.74(4.11)	50.24(18.07)	91.80(4.64)
9	90.10(0.71)	99.48(0.49)	86.39(9.68)	98.29(1.59)	69.26(21.16)	87.96(6.47)	33.06(16.40)	99.84(0.26)	22.87(16.54)	**99.86(0.16)**	76.61(31.68)
OA	74.72(1.98)	86.35(1.28)	87.25(2.84)	92.10(2.02)	89.27(1.67)	91.43(1.49)	87.07(3.83)	85.66(1.06)	81.38(4.95)	83.20(1.51)	**93.17(0.54)**
AA	71.94(0.50)	80.11(1.08)	86.38(2.20)	90.36(1.97)	84.16(3.41)	89.06(3.32)	82.48(5.28)	74.59(1.54)	71.12(7.98)	74.26(1.72)	**91.15(4.43)**
KC	64.89(2.69)	81.60(1.73)	85.04(3.29)	89.49(2.70)	88.49(2.01)	88.49(2.01)	73.52(3.38)	80.83(1.43)	78.00(5.91)	77.71(2.05)	**90.97(0.74)**

competing methods, with smoother results and clearer boundaries, as shown in Fig. 9. This advantage stems from SGDNet's deep exploration of intrinsic structural features within the graph and the optimization of the

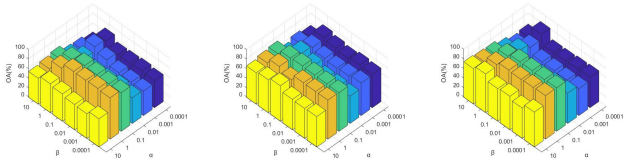
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local feature aggregation process. Compared to traditional GCN-based methods, SGDNet is more flexible in integrating rich contextual information and, through its dynamic subgraph structure adjustment, mitigates the dependency on fixed topological structures, allowing for a more accurate acquisition of nonlinear relationships between land-cover types.

Additionally, it can be observed that the classification maps in Figs. 7–9 of the proposed SGDNet method contain some noise pixels. The causes of this phenomenon can be discussed as follows. During the stage of constructing the initial graph structure, all labeled nodes are extracted and their indices are recorded. Subsequently, these labeled nodes are used to form multiple subgraphs. Subgraph segmentation may cause pixels that were originally adjacent to be separated into different subgraphs, disrupting their spatial continuity. When performing the subgraph training stage, each pixel within a subgraph relies solely on the structural information within the subgraph, neglecting the information aggregation between subgraphs. When predicting the category of pixels within each subgraph, the lack of context information between subgraphs may lead to the presence of noise pixels in classification maps. The DPRN method primarily builds upon CNNs by gradually increasing the feature map dimension at all convolutional layers, which are grouped into pyramidal bottleneck residual blocks, to achieve good HSI classification. The GCN method is a transductive semi-supervised learning approach that aggregates neighborhood feature information through a complete graph convolutional operator instead of a subgraph convolutional operator, thereby preserving good spatial continuity in HSI classification. Based on the above analysis, the main reason for this phenomenon containing a large amount of noise pixels is that the spatial continuity of rich context information is not well maintained in the proposed SGDNet method for HSIs classification. The DPRN and GCN methods are able to preserve the rich context information. However, they are more susceptible to spectral variability, and classification results from both DPRN and GCN methods tend to exhibit large-area errors due to their weak feature extraction ability. Owing to the adaptive graph convolutional advantage of the proposed SGDNet algorithm, which possesses strong feature extraction capabilities and can avoid the impact of spectral variability, large-scale classification errors are effectively mitigated. This further demonstrates that the SGDNet algorithm still holds certain advantages compared to other classification methods.

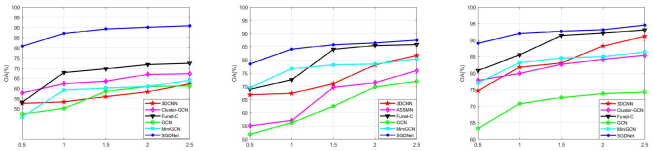
D. Hyperparameter Analysis

In the proposed SGDNet model, there are two primary hyperparameters, α and β, which require adjustment. Therefore, our analysis primarily focuses on assessing the impact of these hyperparameters on the three datasets employed in our study. The experimental results are depicted in Fig. 10. From Fig. 10, we can observe that the classification performance of the SGDNet method exhibits a notable insensitivity to variations in hyperparameters α and β. Specifically, the classification performance remains consistently effective within a range where α and β are not excessively large. This observation underscores the importance of striking a balance between two critical aspects of the model: the embedded features of degree and the aggregated information from GCN in the residuals. Furthermore, it highlights the significance of achieving equilibrium between preserving the initial graph structure and enhancing the graph's structural optimization. It is worth noting that for hyperparameters α and β, optimal results tend to be achieved when these values are set in proximity to 0.2, which appears to be a reliable rule of thumb. This optimal range might be attributed to the notion that it strikes a harmonious equilibrium between the refinement of the initial graph structure and the integration of aggregated information from GCN. This observation suggests that in the adaptive representation process, there is a compelling need to further optimize both the initial graph structure and the inherent information gathered by the GCN.



(a) (b) (c)

Fig. 10. OAs of different tradeoff hyperparameters on the Indian Pines and PaviaU datasets. (a) Indian Pines. (b) Houston. (c) PaviaU.



(a) (b) (c)

Fig. 11. OAs of different training samples for three HSI datasets. (a) Indian Pines. (b) Houston. (c) PaviaU.

E. Analysis of Training Information

To further demonstrate the superiority of SGDNet models with small training sample sizes compared with other methods, OA curves with different numbers of labeled samples on three real HSI datasets, including Indian Pines, Houston, and PaviaU, are shown in Fig. 11. From Fig. 11, the OA of the SGDNet model is superior to that of other methods with different labeled samples. In addition, compared with the GCN method, the SGDNet model can perform graph convolution adaptively, and at the same time, optimize the graph structure information through the output features and the structure of the graph itself, so as to obtain a better convolution effect to adapt to the sample information under different graph structures. Finally, it can be observed that even when the sample size is small, such as 0.5% and 1%, the classification performance of the SGDNet method is greatly superior to the other classification methods. In general, the corresponding classification performance of most popular classification methods is poor due to the limited training samples and large spectral variability in HSI. However, the proposed SGDNet method can deal with this situation, which mainly relies on the cooperation of adaptive convolution and graph optimization. When obtaining the feature information and graph structure

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TABLE V
OAs (%) OF DIFFERENT METHODS ON THREE DATASETS

	Indian Pines	PaviaU	Houston
Cluster-GCN (Base)	66.32	81.37	57.10
Base+Ada	87.41	92.10	81.09
Base+Opt	87.29	88.82	80.97
SGDNet	**88.76**	**93.17**	**83.54**

information output by the initial graph structure, it can capture the similar samples within subgraphs and rebuild the intraclass sample connections via subgraph optimization, improving the classification accuracy.

F. Ablation Study

The SGDNet model contains three key parts to improve classification performance, namely adaptive iteration strategy, embedding feature of integration degree, and subgraph optimization. Here, we use three datasets, Indian Pines, PaviaU, and Houston, to demonstrate the validity of these three operations. Set the number of training samples for each class on these three datasets to 1.5%, 1%, and 0.5% of the current class, respectively. The experimental results are shown in Table V. Base+Ada denotes that the Cluster-GCN method is with the proposed subgraph-dependent convolution. Base+Opt denotes that the Cluster-GCN method is with the proposed subgraph optimization. Finally, SGDNet is the proposed method. The experimental results show that the classification performance of the Base+Ada method is better than Cluster-GCN, which proves the effectiveness of the adaptive iterative optimization strategy. Second, the classification performance of the Base+Opt method is better than the Cluster-GCN method because the subgraph optimization plays a crucial role in the expression of graph structure information in HSI classification. Finally, the classification performance of the SGDNet method is better than the other methods due to the effective fusion between subgraph-dependent convolution and subgraph optimization modules.

G. Time Complexity Analysis

The time complexity analysis of the proposed SGDNet method encompasses the following several aspects. First, the complexity of the similarity matrix construction is $[O(n^2 \cdot d)]$, where n denotes the number of nodes within the subgraph and d represents the dimensionality of the features. Subsequently, as for the optimization of the subgraph structure, Dijkstra's algorithm is employed to calculate the shortest paths from a single source to all other vertices in a weighted graph. This step is computationally intensive, with the time complexity $O(n^2)$. Then, the integration of positional encoding with the cosine similarity of node features enriches the construction of the similarity matrix. This step also involves linear transformations and bears the time complexity $O(n \cdot d)$. After obtaining the similarity matrix, updating the similarity matrix and the forward propagation through the message-passing mechanism are performed with the time complexity $O(nd^2)$. Finally, the computation of intraclass distances, primarily based on the Euclidean distances between nodes, has the time complexity $O(|c_i| \cdot d)$, with $|c_i|$ indicating the number of nodes within class i. Given that the number of classes is typically a constant, this step's time complexity is relatively minor in the overall scheme. In summary, the time complexity for training a single subgraph is predominantly governed by the $O(n^2)$, which is primarily from the computational cost associated with the similarity matrix construction and shortest path calculations. Additionally, the running time of the proposed SGDNet algorithm and other methods is also presented in Table VI. From Table VI, it can be observed that the running time of the SGDNet algorithm on the three datasets is relatively longer compared to other algorithms. This is due to the fact that the complexity of the SGDNet algorithm is approximately $O(n^2)$. Despite the longer running time, it remains within an acceptable range. Moreover, the classification accuracy of the proposed SGDNet algorithm is superior to that of other algorithms. Therefore, the proposed SGDNet algorithm achieves a favorable balance between running time and classification accuracy.

TABLE VI
RUNNING TIME(S) FOR DIFFERENT METHODS ON THREE DATASETS

Methods	Indian Pines	Houston	PaviaU
3DCNN	2	3	2
DPRN	119	122	141
ASSMN	59	59	255
DSGSF	69	344	49
morphFormer	163	286	518
GraphGST	4	6	4
GCN	41	18	78
Funet-C	41	53	82
Cluster-GCN	3	2	4
MiniGCN	49	62	82
SGDNet	410	532	701

H. Visualization Effects of Subgraph Optimization

The visualization effects of subgraph optimization on three HSIs dataset, including Indian Pines, PaviaU, and Houston, can be found in Fig. 12, where each number denotes a node in the graph. Different node colors indicate different categories. Loops on a single node represent self-loops. Prior to subgraph optimization, the initial subgraphs generated by the segmentation algorithm exhibit substantial isolated nodes, as demonstrated in Fig. 12(a), (c), and (e). These isolated nodes struggle to form cohesive homogeneous cluster structures with nodes of identical categories, which compromises both intraclass feature smoothness and interclass discriminability. Notably, even within categories containing abundant samples, isolated nodes persist due to insufficient node connectivity. This structural deficiency severely impedes information propagation between nodes and consequently degrades SGNN performance. Given that SGNNs fundamentally rely on node connectivity for message passing and feature aggregation, the presence of isolated nodes significantly undermines the efficacy of the message-passing mechanism. Following subgraph optimization, the subgraph topology achieves marked

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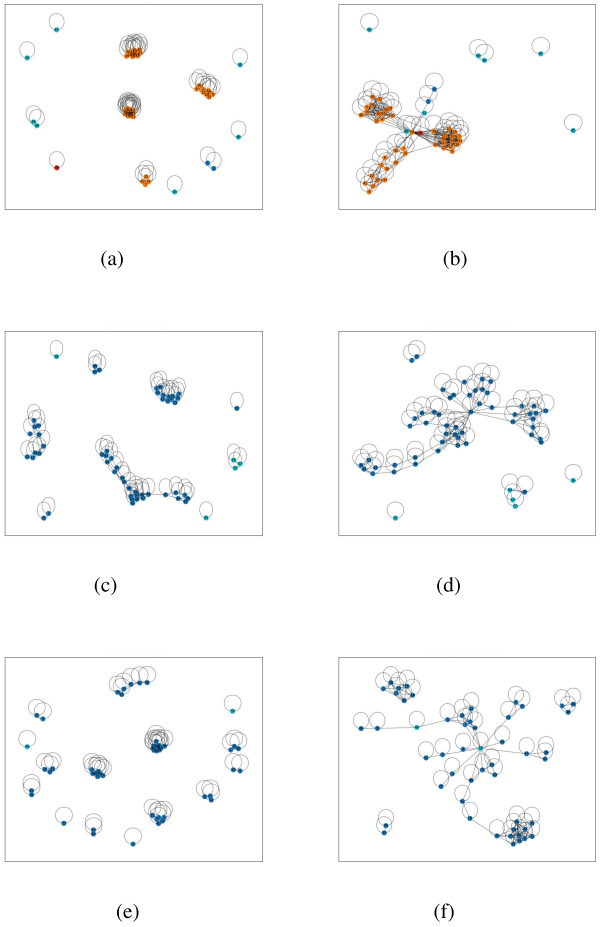


Fig. 12. Visual comparison between initial and optimized subgraphs. (a) Initial subgraph in Indian Pines. (b) Optimized subgraph in Indian Pines. (c) Initial subgraph in PaviaU. (d) Optimized subgraph in PaviaU. (e) Initial subgraph in Houston. (f) Optimized subgraph in Houston.

structural refinement as shown in Fig. 12(b), (d), and (f). The optimized configuration substantially reduces both isolated nodes and fragmented clusters while establishing effective cluster connections. Although residual isolated nodes may persist in rare cases due to extreme adjacency sparsity, the majority exhibit enhanced connectivity. The restructured subgraph significantly strengthens intraclass information propagation capabilities, thereby improving model performance in category-specific recognition and classification tasks.

V. Conclusion

In order to address the aggregation strategy of fixed K-order neighbors' information within an imbalanced subgraph structure in the message-passing mechanism, we proposed a novel SGDNet method that can adaptively capture the neighbor orders of each subgraph, achieving flexible feature information aggregation and good HSI classification results. Specifically, the designed subgraph-dependent convolution consists of degree feature embedding with the residual, and the intraclass distance within the subgraph determines the optimal subgraph convolution layers for imbalanced subgraph structure in a feedback way. Then, the designed strategy of structural long-range dependency using an anchor point-based position coding within the graph effectively captures the long-range structural information with the help of the designed structural loss function, achieving effective subgraph optimization. Comprehensive experiments on three popular HSI datasets, including Indian Pines, Houston, and PaviaU, demonstrated the superior performance of the SGDNet model, compared to other popular classification methods.

Despite the good results provided by the proposed SGDNet, there are several unresolved issues that may present challenges over time. In particular, our future work will be aimed at the following directions: 1) designing the strategy of information aggregation between different subgraphs to achieve effective message passing from richer contextual information and 2) reducing the computational complexity of the proposed SGDNet by developing new methods.

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